

Lab 3 – Architecture Selection & Justification

Faculty Research Grant & Publication Tracker System | SRN: PES1UG24AM076

Selected Architecture

A Layered (3-tier) Architecture was selected for this system.

The design is split into a Presentation Tier (Researcher Portal and Dean Dashboard), an Application/Service Tier (Grant Service, Publication Service, Co-author Approval Service and Reporting Service) and a Persistence Tier (Data Access Layer, Research Database and Audit Trail Store). Requests flow strictly top-down: the UI calls the services through IGrantAPI, IPublicationAPI, IApprovalAPI and IReportAPI, and the services reach storage only through the IRepository interface.

Why Layered Architecture Fits This Scenario

- 1. Two user roles, one shared set of rules.** Researchers and the Dean have very different needs, so they get separate front-ends, but both rely on the same grant, expense and publication data. Placing all logic in a shared service tier lets two different user interfaces reuse one set of rules instead of duplicating them, and either interface can be redesigned without touching the backend.
- 2. Easy to maintain as rules change.** Requirements such as "an expense cannot exceed the grant limit" and "a publication needs co-author approval" change over time as university policy changes. Because each concern lives in its own service in a single tier, a policy change is made in one component and tested in isolation. A microservices design would be excessive for a single-department tool and would add deployment and data-consistency overhead the team does not need.

Security Advantage

Each tier is a security boundary. Users can only reach the service tier over HTTPS, where login and role checks are applied (for example, only the Dean can open department-wide reports). The services never build raw SQL; the Data Access Layer uses prepared statements, which blocks SQL injection. Every change to an approval or a financial record is also written through IAuditTrail into an append-only, hash-linked Audit Trail Store, so records cannot be edited or erased without detection, meeting NFR-001.

Performance Benefit

Separating tiers allows each one to be optimised independently. The Reporting Service can pre-compute and cache fund burn-up figures instead of recalculating them on every dashboard load, and the stateless service tier can be run as multiple instances behind a load balancer when many faculty submit reports near a deadline. The Data Access Layer adds connection pooling and indexing on frequently queried fields (grant ID, faculty ID), keeping response times within the limits of NFR-002.